

VASU SHARMA

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Software Developer | Data Engineer | Data Analyst

PROFESSIONAL EXPERIENCE

Experienced professional with 5+ years of hands-on work experience in **Python, database management and cloud technologies**, complemented by **BS in Computer Science & MS in Data Science**. Seeking opportunities in roles **Software Development, Data Engineering, or Data Analysis**.

WORK EXPERIENCE

Indiana University Observatory on Social Media (OSoMe)

Full Stack Developer

Bloomington, IN

Jan 2024 - May 2024

- Designed and implemented an annotation portal utilizing **Django Forms, HTML, CSS, and JavaScript**, for **live data** to improve **accuracy of annotation process by 70% and** reducing errors by **45%**.
- Achieved over **30%** optimization enhancement in the data analysis pipeline by standardizing the format of results produced by the portal.
- Designed PGSQL database schema, leveraging Object-Relational Mapping (ORM) for efficient **CRUD operations** between frontend and backend.

Indiana University School of Public Health

Research Assistant

Bloomington, IN

Jan 2024 - May 2024

- Created **python script** scheduled via shell script extracting over **4 TB** data from Women's Health Initiative Server.
- Constructed **ETL python pipeline** for feature extraction, transforming, and applying ML algorithms.

Good United

Python Developer; Company: Tatra Data Services Private Limited, Delhi, India

Charleston, SC

Feb 2022 – Aug 2022

- Engineered **Python-based bot** with **serverless AWS architecture**, optimizing **user data extraction** for Facebook campaigns by **40%**.
- Streamlined **API response time by 45%** after integrating **AWS Elastic Cache** into **Flask APIs**, to handle increased user load on the system.
- Developed **Python synchronization Cron jobs** in AWS for table **data denormalization**, reducing data load time by **70%**.
- Executed **indexing and partitioning** strategies, reducing **CPU usage** on production servers by **60%** and optimizing data reload time for **materialized views by 85%**.

Globalshala

Data Analyst Intern

Remote

Jan 2022 - Feb 2022

- Spearheaded analysis of Facebook ad campaigns based on parameters such as **Reach, Unique Clicks, Revenue, Cost per Clicks**.
- Built advanced database queries with visualization in **Tableau and Power BI** for better decision making and ROI.

Capital Group | American Funds

Senior System Engineer; Company: Infosys Ltd., Pune, India

Irvine, CA

Aug 2017 – Feb 2021

- Played pivotal role in **designing and implementing a data model for trade processing application**, leveraging step functions to produce **SWIFT instructions** for **custodian banks** and generate **critical reports** for business users, in a team of 10 members.
- Led team of **4 members** to establish a custom workflow for generating SWIFT instructions for US-based trades, utilizing **Flask, Django APIs, Python scripts, and database elements like stored procedures, tables, and triggers**, increasing client revenue by **more than 60%**.
- Formulated **Python ETL workflows** to extract trade summaries from upstream applications, optimizing data transfer and processing by 30%.
- Optimized data transfer between upstream and downstream by **40%** by implementing **Python APIs**.
- Released **Python and PowerShell scripts** to automate **report generation and service status monitoring**, streamlining processes, reducing **redundancy by 30%**.

Infosys Ltd.

Software Developer Intern

Mysore, India

Jan 2017 - May 2017

- Trained on Python, SQL, Flask, Django.
- Supervised 6 members to construct an online shopping Portal using **Django, SQL, HTML, CSS**.

ACADEMIC PROJECTS

Sentiment Analysis | Skills : Pandas, FLASK, HTML, CSS, SCIKIT | [LINK](#)

Nov 2023 - Dec 2023

- Performed Data cleaning, feature extraction, EDA on more than 20GB dataset obtained from E-Commerce website to analyze user review sentiment.
- Enhanced ML algorithm performance by **40%** after performing **Hyperparameter tuning**. Deployed model on **Flask website**.

Salary Predictor | Skills : Pandas, FLASK, HTML, CSS, SCIKIT | [LINK](#)

Jul 2023 - Aug 2023

- Evaluated salary of IT professionals using ML algorithms. Model was rolled out using the **Flask website with 80% accuracy**.

Solar Energy Predictor | Skills: Pandas, Django, HTML, CSS, SCIKIT | [LINK](#)

Mar 2023 - Apr 2023

- Predicted Solar Energy Production for Europe using Time Series Models and released **Django website with 85% accuracy**.

TECHNICAL SKILLS

- **Programming** : Python (Pandas, NumPy, Pyspark, TensorFlow, Scikit, NLTK, Seaborn), PowerShell, C# , C, R, Linux, HTML5
- **Databases/Cloud** : PostgreSQL, Sybase, Neo4j, SQL, Amazon Redshift, Mongo DB
- **Cloud Technologies** : AWS (SQS,RDS, DynamoDB, EC2, S3, Lambda,), GCP(Big Query, Looker)
- **Frameworks** : Flask, Django, Streamlit
- **Other Technologies** : Apache Spark, POSTMAN, Autosys, JIRA, XL Deploy, Bitbucket, Bamboo, Version Control, Tableau

EDUCATION

Indiana University Bloomington,

Bloomington, IN

- Master of Science in Data Science

May 2024

SRM University

Delhi, India

- Bachelor of Technology in Computer Science and Engineering

May 2017

Certifications: Big Data with **Apache Spark and Python**, Data Engineering using **AWS Data Analytics**, **Python for Data Science** from UDEMY